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July 27, 1973
SUPERSEDING
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July 26, 1967

FEDERAL SPECIFICATION

GRINDERS, MEAT, ELECTRIC; GRINDERS, MEAT, HAND; AND MIXER-GRINDER, MEAT, ELECTRIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand-operated and electric motordriven meat grinders (commercial designation "chopper") and electric motordriven meat mixer-grinder.

1.2 Classification. The grinders shall be of the following items, types, sizes, and styles, and sizes of mixer-grinders as specified (see 6.2).

Item 1	-	Grinders.
Type I	-	Hand-operated (for bench or table mounting).
Size 3	-	2-1/2 pounds capacity*.
Size 5	-	5 pounds capacity.
Size 8	-	8 pounds capacity.
Size 10	-	10 pounds capacity.
Type II	-	Motor operated.
Size 4	-	4 pounds capacity**.
Style 1	-	For bench or table use (portable).
Size 8	-	8 pounds capacity.
Style 3	-	For permanent bench or table mounting.
Size 16	-	16 pounds capacity.
Size 20	-	20 pounds capacity.
Size 24	-	24 pounds capacity.
Style 2	-	Floor mounted.
Style 3	-	For permanent bench or table mounting.
Size 30	-	30 pounds capacity.
Size 50	-	50 pounds capacity.
Style 2	-	Floor mounted.

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Item 2	-	Mixer-grinder, motor operated.
Size 40	-	40 pounds capacity.
Size 60	-	60 pounds capacity.
Size 100	-	100 pounds capacity.
Item 3	-	Mixer-grinder, motor operated, continuous ground meat system, 100 pound capacity** (2 mixer-grinder integrated system).
Item 4	-	Chopping heads.
Size 42		
Size 46		
Size 52		

*Capacity of item I, type 1 grinders is defined as the amount of meat processed in 100 revolutions of the crank using 3/16-inch hole and end-plate.

**Capacity of item 1, type II grinders, item 2, and item 3 mixer-grinders are defined as the amount of meat processed in one minute using a 1/8-inch hole-plate.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CC-M-636	-	Motors, Alternating-Current, (Fractional Horsepower).
CC-M-641	-	Motors, Alternating-Current, (Integral Horsepower, 200 HP and Smaller).
QQ-I-666	-	Iron, Malleable, Ferritic for Castings.
QQ-S-681	-	Steel Castings.
QQ-S-698	-	Steel, Sheet and Strip, Low Carbon.
QQ-S-700	-	Steel Sheet and Strip, Medium and High Carbon.
QQ-S-741	-	Steel, Carbon: Structural Shapes, Plates, and Bars.
QQ-S-763	-	Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.
MMM-A-260	-	Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
PPP-B-576	-	Boxes, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-591	-	Boxes, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.

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PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-C-104 - Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-B-3854 - Beef, Ground, Frozen.
MIL-L-10547 - Liners, Case and Sheet, Overwrap, Water-Vapor or Waterproof, Flexible.
MIL-I-11466 - Iron, Nodular Graphitic (Ductile Iron) Castings.
MIL-P-15024 - Plate, Tags and Bands for Identification of Equipment.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirement for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specificatir to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

A 48 - Gray Iron Castings.
A 120 - Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Pipe for Ordinary Uses.
A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip.
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Electric Manufacturers' Association (NEMS) Publication:

MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 115 East 44th St., New York, NY 10017.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago. UL 60606.)

Underwriters' Laboratories (UL), Inc. Standard:

No. 73 - Motor-Operated Appliances.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Rd., Melville, NY 11746; 207 East Ohio St., Chicago, IL 60611; or 1655 Scott Blvd., Santa Clara, CA 95050.)

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National Sanitation Foundation (NSF) Standards:

No. C-2 - Basic Criteria for the Evaluation of Special Equipment and/or Devices
No. 8 - Commercial Powered Food Preparation Equipment
Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48106.)

(Technical society and technical association specifications and standards are generally available from reference libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample meat grinder or sample mixer-grinder for first article inspection and approval (see 4.3 and 6.4).

3.2 Standard product. The meat grinders and mixer-grinders delivered under this specification shall be the manufacturers standard commercial product except for any changes necessary to comply with specification requirements. All meat grinders of one type, style and size and all mixer-grinders furnished under this specification for any one contract, including parts and assemblies thereof, shall be interchangeable.

3.3 Standards. The electric meat grinders and mixer-grinders shall comply with the applicable requirements of NSF Std. No. 8 and UL Std. No. 73.

3.3.1 Compliance. Prior to approval of the first article if one is submitted, or prior to commencing production the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the electric meat grinders and mixer-grinder he proposes to furnish under this specification meet the applicable requirements of NSF Std. No. 8 and UL Std. No. 73 as follows:

3.3.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF Seal on the finished meat grinder or mixer-grinder or

(2) A certification for the meat grinder or mixer-grinder issued by NSF under their special one-time evaluation/certification service shall be submitted prior to approval of the first article, if one is submitted, or

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(3) A certified test report from a recognized independent testing laboratory acceptable to the medical authority functioning in an advisory capacity to the preparing agency of the specification indicating that the item 1, type I meat grinder and item 1, type II meat grinder or items 2 and 3 mixer-grinder have been tested and conform to NSF Std. No. C-2 and No. 8, respectively.

3.3.1.2 UL. Acceptable evidence of meeting the applicable requirements of UL shall be the UL Label, UL Listing Mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the meat grinder or the mixer-grinder, as applicable, have been tested and conform to the applicable requirements of UL 73.

3.4 Materials and components. Material and components shall be as specified herein: Materials not definitely specified shall be of the quality normally used by the manufacturer on his standard meat grinders and mixer-grinders provided the completed items comply with all provisions of this specification.

3.4.1 Stainless steel.

3.4.1.1 Stainless steel plates, sheets, or strips. Stainless steel plates, sheets, or strips shall conform to type 302 or 304 of ASTM A 167 and type 410 or 430 of ASTM A 176, as applicable.

3.4.1.2 Bar stock. Corrosion-resistant steel bar stock shall conform to class 302 or 304 cold finish of QQ-S-763.

3.4.2 Castings.

3.4.2.1 Steel. Steel castings shall be any class from 80-50 to 175-145, annealed condition of QQ-S-681.

3.4.2.2 Cast iron. Cast iron castings shall conform to ASTM A-48.

3.4.2.3 Malleable iron. Malleable iron castings shall be grade II of QQ-I-666.

3.4.2.4 Ductile iron. Ductile iron castings shall be grade D4512 or D5506 of MIL-I-11466.

3.4.3 High carbon steel. High carbon steel shall conform to QQ-S-700.

3.4.4 Low carbon steel. Low carbon steel shall conform to any steel number from 1009 through 1020 of QQ-S-698.

3.4.5 Angles and shapes. Steel angles and shapes shall conform to QQ-S-741.

3.4.6 Pipe. Pipe shall conform to ASTM A 120.

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3.5 General requirements. General requirements for item 1 grinder and item 2 mixer-grinder are as follows:

3.5.1 Feed worm housing. Feed worm housings for item 1 grinders and item 2 mixer-grinders shall be one integral casting of the material specified in 3.4.2.1, 3.4.2.2, or 3.4.2.3. The inside diameter of the feed worm section shall be provided with flutes. Flutes shall be sized so as to provide a uniform clearance with the worm.

3.5.2 Feed worm. Feed worms for item 1 grinders and item 2 mixer-grinders shall be cast of the material specified in 3.4.2.1, 3.4.2.2, 3.4.2.3, or 3.4.2.4. Bearing surfaces shall be smoothly machined. The worm and end-plate bearing shall be cast in one piece or the worm may be provided with a threaded steel stud for an end-plate bearing. The steel stud shall be provided with a square or hexagonal section to allow insertion and removal with a wrench. The feed worm shall be provided with a spiral flute and shall be removable from the grinder body.

3.5.3 Knives. Knives for item 1, type I grinders shall be high carbon steel specified in 3.4.3, low carbon steel specified in 3.4.4, malleable iron specified in 3.4.2.3 or forged steel. Knives for item 1, type II grinders and item 2 and item 3 mixer-grinders shall be high carbon steel specified in 3.4.3 or forged steel. Knives shall be removable and shall have not less than four blades with sharpened cutting edges. The surface of the blades designed to come in contact with the end plate shall be ground flat. The blades for item 1, type I grinders shall have a through-hardened or carburized Rockwell "C" hardness not less than 45 nor more than 60 when tested as specified in 4.6.2. The blades for item 1, type II grinders and item 2 and item 3 mixer-grinders shall have a through-Rockwell "C" hardness not less than 45 nor more than 60 when tested as specified in 4.6.2.

3.5.4 End-plates. End-plates for item 1 grinders, item 2 and item 3 mixer-grinders, and item 4 end-plates (see 1.2) shall be high carbon steel specified in 3.4.3, low carbon steel specified in 3.4.4, or 410 stainless steel specified in 3.4.1.1. End-plates may be provided with or without hubs. End-plates shall have a carburized or through-Rockwell "C" hardness not less than 45 nor more than 60 between perforations when tested as specified in 4.6.2. The plates shall be notched so as to interlock with a pin in the housing, or one or more flat(s) shall be provided on the diameter to fit a corresponding segment cast into the housing. Both designs shall prevent rotation of the end-plate during operation.

3.5.5 End-plate clamping ring. Each item 1 grinder and each item 2 and item 3 mixer-grinder shall be provided with a threaded clamping ring for positioning the perforated end-plate against the knife blade. The ring shall be provided with recesses, lugs, or handgrips for tightening and removal. Material for the ring shall be the same as that for the housing.

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3.5.6 Thrust bearing. Item 1, type I grinders may be provided with or without a thrust bearing. When a thrust bearing is not provided, smoothly finished bearing surfaces shall be provided on the worm and the member against which the thrust is impressed. Item 1, type I grinders with a thrust bearing item 1, type II grinders and item 2 and item 3 mixer-grinders shall have a bearing located between the feed worm and the member against which the thrust is impressed. The bearing shall be a flat metal ring, a roller-or ball-bearing, or a hard-bone-fibre disk. Bearings and disks shall be removable for replacement.

3.5.7 Interchangeability. All replaceable parts shall be constructed to definite standards, tolerances, and clearances so that any part of a particular type, size or style may be replaced without requiring modification.

3.6 Detailed requirements.

3.6.1 Item 1, type I. Item 1, type I grinders shall consist of a feed worm housing, bowl, a feed worm to force meat against the blades of the cutter, a perforated end-plate, a handcrank, and a clamping device. The grinders shall be so constructed that all parts coming in contact with the product being ground can be easily removed for cleaning and sanitizing, or so designed that adequate cleaning and sanitizing can be accomplished in place.

3.6.1.1 Knife. The knife shall conform to the requirements of 3.5.3. Unless otherwise specified (see 6.2), knives shall be furnished in the quantity shown in table I.

3.6.1.2 End-plates. End plates shall conform to the requirements of 3.5.4. Unless otherwise specified (see 6.2), end-plates shall be furnished in the quantity and with the size perforations shown in table I.

TABLE I. Item 1, type I (knives and end-plates)

Grinder size	Quantity of knives per grinder	Quantity of end-plates per grinder			
		3/16-inch holes	1/4-inch holes	3/8-inch holes	1/2-inch holes
3*	3	1	1	1	-
5	3	1	-	1	1
8	3	1	-	1	1
10	3	1	-	1	1

*When end-plates having elliptical slots or holes are required (see 6.2), for size 3 grinder, the quantity shall be specified.

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3.6.1.3 Base. Unless otherwise specified (see 6.2) the item 1, grinder shall be provided with a screw-clamp base. When specified, a slide-clamp base or a screwdown base shall be provided. Screw clamp bases shall have one or more threaded steel screws with a wing type handgrip and either a pivoted-bearing button or pivoted-bearing bar. The screw-clamp base shall be suitable for clamping to a table or bench up to 1-1/2 inches thick. Slide-clamp and screw-down bases shall be suitable for permanent attachment by screws to either a bench or table. When tested as specified in 4.6.1, the screw clamp and grinder body shall show no deformation.

3.6.1.4 Handcrank. The handcrank shall be cast of the material specified in 3.4.2.1, 3.4.2.3, or 3.4.2.4. The crank shall be detachable and shall be provided with a free-turning wood, plastic, or metal handle grip.

3.6.1.5 Dimensions. The item 1, type I grinders shall conform to the dimensional requirements shown in table II.

TABLE II. Item 1, type I dimensions

Machine	Minimum outside length of body exclusive of boss at crank end	Minimum outside diameter of body	Minimum hopper outside dimensions at top	End plates	
				Minimum outside diameter	Minimum thickness
	Inches	Inches	Inches	Inches	Inch
3	3-1/2	2-1/8	3 by 3-7/8	2-1/16	1/8
5	6	2-5/8	4 by 4-3/4	2-11/16	3/16
8	7-1/4	3-3/16	4-7/8 by 5-7/8	3-1/4	1/4
10	8-3/4	3-7/8	5 by 7	3-7/8	9/32

3.6.1.6 Capacity. The sizes 3, 5, 8, and 10 grinders shall be capable of grinding 2-1/2, 5-, 8-, and 10- pounds, respectively, of medium grade raw beef per one hundred crank revolutions when tested as specified in 4.6.3.1. The ground meat shall show no discoloration indicating excessive pressure.

3.6.1.7 Finish. Burrs, slivers, and sharp and rough edges shall be removed. Metal surfaces of the item 1, type I grinders except bearing surfaces, end-plates, cutting edges, and edges of worm flutes shall be uniformly tincoated in accordance with standard commercial practice. The coating shall be adherent and free from blisters, pits, or nodules.

3.6.2 Item 1, type II. Item 1, type II grinders shall consist of a base-plate, housing, electric motor, transmission, bowl of feed chute, feed worm housing, clamping ring, feed worm, end plates, and meat pan with guard. Styles 1 and 3 grinders shall be for countertop use. Style 1 shall be portable and style 3 shall be for permanent bolting to a bench or table. Style 2, shall be for floor mounting.

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Base plates shall be cast of the material specified in 3.4.2.1, 3.4.2.3, 3.4.2.4, or fabricated from carbon steel specified in 3.4.4. All fitted surfaces shall be smooth and properly aligned. The grinder shall be assembled so the assembly of component parts is complete and of proper interrelationship, and when tested as specified in 4.6.3.2, moving parts shall show no undue vibration, bearings shall not overheat, and controls shall function as intended.

3.6.2.1 Base plate, styles 1 and 3. The base plate for style 1 grinders shall have four legs or rubber supports or may have a rubber gasket around the entire outside edge of the base. The base plate for the style 3 grinders shall have four legs or shall be designed for sealing to a bench or table top.

3.6.2.2 Base plates, style 2. Base plates for style 2 grinders shall be provided with four stainless steel, aluminum, or galvanized or nickel plated steel legs. Each leg shall be provided with an adjustable foot or three legs shall be a fixed length with one leg adjustable for leveling. When specified (see 6.2), feet shall have provisions for bolting to the floor. Unless otherwise specified (see 6.2) height from the floor to the center of the end-plate shall be 24 $\pm$ 1 inch except sizes 20, 24, 30, and 50 grinders shall be 28 inches $\pm$ 1 inch.

3.6.2.3 Feed worm housing. The feed worm housing shall conform to the requirements of 3.5.1, and shall be detachable from the transmission housing. The bowl or feed chute from the meat pan to the feed worm shall be cast integral with the feed worm housing or shall be removable. The removable feed chute shall be fabricated of aluminum.

3.6.2.4 Knife. The knife shall conform to the requirements of 3.5.3. Unless otherwise specified (see 6.2), knives shall be provided in the quantity shown in table III.

3.6.2.5 End-plates and clamping ring. End-plates and clamping ring shall conform to the requirements of 3.5.4 and 3.5.5, respectively. Unless otherwise specified (see 6.2), end-plates shall be furnished in the quantity and with the size perforations shown in table III.

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TABLE III. Item 1, type II (knives and end-plates)

Grinder size	Quantity of knives per grinder	Quantity of end-plates per grinder							End-plate thickness (minimum), inch	End-plate outside diameter (minimum), inches
		1/8-inch holes	3/16-inch holes	1/4-inch holes	3/8-inch holes	1/2-inch holes	3/4-inch holes	1-inch holes		
4	3	1	1	1	1	-	-	-	7/32	2-3/4
8	3	1	1	1	1	-	-	-	1/4	2-3/4
16 1/	3	1	1	1	1	1	-	-	9/32	3-7/32
20	3	1	1	1	1	1	1	-	9/32	3-13/16
24	3	1	1	1	1	1	1	-	11/32	3-7/8
30	3	1	1	1	1	1	1	-	13/32	3-7/8
50	3	1	1	1	1	1	1	1	13/32	5

1/ The 1/2-inch hole plate is not required on size 16 grinders for shipboard use.

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3.6.2.6 Meat pan. The meat pan shall be stainless steel. The pan capacity shall be that normally furnished by the manufacturer on his standard product of equal capacity except size 30 and 50 grinders shall have a high back and shall hold 100 and 150 pounds of meat, respectively. The pan shall be provided with a hole to the feed chute or bowl and shall be supported to fit in proper position over the feed chute or bowl. One or more safety guards shall be secured to the pan in a location that will prevent attendant's fingers from accidentally contacting the feed worm or the size and length of the feed opening shall be of such size as to prevent operator's fingers from contacting feed screw.

3.6.2.7 Stomper. A meat stomper shall be provided with each grinder. The stomper shall be a lever operated design or a separate item. When a separate item is provided, it shall be plastic or aluminum and shall be sized and shaped to prevent contact with the feed worm when inserted into the bowl.

3.6.2.8 Power transmission. The grinder shall be equipped with necessary gears, gear shafts, chains, and sprockets for transmission power from the motor shaft to the feed worm. Components shall be encased and securely fastened within the housing. Power transmission of size 50 grinders shall be either a single reduction type on a double reduction type. The double reduction type shall be either a separate reduction unit or a combination motor and reduction unit.

3.6.2.8.1 Bearings. Transmission bearings shall be ball, roller, needle, or sleeve type. Sleeve bearings shall be renewable and shall be bronze or other suitable bearing metal. Bearings shall be located and arranged to prevent lubricating oil or grease from contacting meat.

3.6.2.8.2 Gears, sprockets, and chains. Gears, sprockets, and chains shall be of adequate load-bearing capacity to meet the tests specified in 4.6.3. Gear teeth shall be machine cut and may be spur, helical, or worm. Gears, shafts, and sprockets shall be alloy steel except a composition gear is permissible for size 24 and smaller machines on the first speed reduction. When worm gearing is used, the worm wheel may be bronze. Gears and sprockets shall be keyed or splined to the shafts.

3.6.2.9 Lubrication. Means shall be provided for lubrication of bearings, gearing, and other parts requiring lubrication. Encased gears shall operate in an oil bath. An oil level indicator or removable cap shall be furnished when gears are encased in oil.

3.6.2.10 Capacity. Item 1, type II grinders shall be capable of grinding the quantity of meat specified in table IV when tested as specified in 4.6.3.2. Capacity shall be based on the second grinding of medium grade raw beef. The ground meat shall conform to the requirements of type I of MIL-B-3854.

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TABLE IV. Capacity, item 1, type II

Grinder size	Minimum rate through 1/8-inch hole and end-plate for all direct current and 60-cycle* alternating current grinders
	Pounds per minute on second grind
4	4
8	8
16	16
20	20
24	24
30	30
50	50

*For grinders with 25- or 50 cycle alternating current motors, the rate shall be not less than 80 percent of the values specified in table IV.

3.6.2.11 Motors. Motors, shall be either alternating current (a.c.) or direct current (d.c.), as required. Alternating current motors shall comply with CC-M-636 or CC-M-641, as applicable. Direct current motors shall be compound wound or stabilized shunt wound. The motor operating voltage and the type current supply (if a.c., the phase and frequency) shall be as specified (see 6.2). Single-phase a.c. motors provided on size 4 grinders shall be repulsion-induction capacitor, repulsion-start induction, or capacitor-start. Single-phase a.c. motors for other sizes of grinders shall be capacitor, capacitor-start, repulsion-induction, or repulsion-start-induction. Three-phase motors shall be squirrel-cage type. Motors shall have the following characteristics:

Ambient temperature rating	40° C.
Enclosure	Totally enclosed, or drip proof.
Horsepower	As specified in 3.6.2.12.
Speed	Constant.
Duty	Continuous.
Bearings	Ball or sleeve.
Insulation	Class A or B.
Temperature rise	NEMA* standard for the submitted enclosure and insulation.

*Consult NEMA Std. MG1.

3.6.2.12 Motor horsepower. Motors shall have the following minimum horsepower (h.p.) ratings:

Size	Horsepower
4	1/4
8	1/2
16	1
20	2
24	2
30	3
50	7-1/2

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3.6.2.13 Motor controls.

3.6.2.13.1 Direct current and single-phase, alternating current. For d.c. and single-phase a.c., the switches, controllers, and starters shall conform to the applicable subparagraph of 3.6.2.13 and shall be provided as specified in table V.

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TABLE V. Motor switches and starters for d.c. and single-phase a.c.

Grinder size	115-volt motors	208-and 230-volt motors
4	a.c. - Motor circuit switch.	a.c. and d.c. - Motor circuit switch.
8	a.c. - Motor circuit switch. d.c. - Either manually or magnetically operated starter and one or more steps of armature starting resistance.	a.c. and d.c. - Motor circuit switch.
16	a.c. - Motor circuit switch. d.c. - Either manually or magnetically operated starter and two or more steps of armature starting resistance.	a.c. - Motor circuit switch. d.c. - Either manually or magnetically operated starter and one or more steps of armature starting resistance.
20		a.c. - Motor circuit switch. d.c. - Either manually or magnetically operated starter and one or more steps of armature starting resistance.
24 and 30		a.c. - Motor circuit switch. d.c. - Either manually or magnetically operated starter and one or more steps of armature starting resistance.
50		a.c. - Magnetic across-the-line contractor breaking all incoming lines with start-and-stop pushbutton control having undervoltage protection. d.c. - Manually operated armature resistor type, reduced voltage starter having undervoltage protection.

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3.6.2.13.2 Three-phase a.c. For three-phase a.c., the grinders shall be equipped with the following:

(a) A motor circuit switch conforming to 3.6.2.13.3.
(b) A magnetically operated starter conforming to 3.6.2.13.4.
(c) Under voltage protection conforming to 3.6.2.13.6 except for polyphase motors.
(d) Overload protection conforming to 3.6.2.13.7.

3.6.2.13.3 Motor circuit switch. The motor switch shall be an enclosed quick-break, externally operated, across-the-line, indicating type. The switch shall be of the recess type and shall open all leads.

3.6.2.13.4 Starters. Starters shall be manually or magnetically operated as required by table V and shall be enclosed. Manual starters shall have an externally operated lever or pushbutton. Magnetic starters shall have an external pushbutton of the recessed type.

3.6.2.13.5 Reduced-voltage starters. When manually operated, reduced-voltage starters are provided, the starters shall be enclosed and provided with undervoltage protection.

3.6.2.13.6 Under-voltage protection. Under-voltage protective devices shall cause and maintain interruption of power in the supply circuit when a reduction of 10% or more from the specified voltage occurs.

3.6.2.13.7 Overload protection. Each motor shall be provided with a device for over-current protection. The device shall be a thermal cut-out of the inverse time delay, arranged for automatic or manual reset.

3.6.2.14 Wiring. Unless otherwise specified (see 6.2), all item 1, type II grinders except sizes 4 and 8 shall be for permanent connection to electric supply. Starting switches and controls shall be attached to the grinder and wired to the motor. All power supply connections shall provide grounding of the grinder. The wired machine shall be capable of withstanding the di-electric test specified in 4.6.5 without electrical breakdown. Sizes 4 and 8 grinders shall be provided with a 3-wire, 6-foot, flexible, heavy-duty, rubber-jacketed, type S cord with a 3-prong grounding type attachment plug cap.

3.6.2.15 Fungus and moisture-resistance. When specified (see 6.2), components of the motor and other electrical equipment shall be treated to resist moisture and growth of fungus as follows:

a. Circuit elements which have a temperature rise of not more than 75 F. when operating at full load, shall be coated with fungus-resistant varnish conforming to MIL-V-173. Circuit elements include, but are not limited to, cables, wire, terminal blocks, capacitors, and coils.

b. Electrical components such as switches, fuses, and contacts, shall not be treated. Other materials and components which are inherently fungus resistant or are protected by hermemetic sealing need not be treated.

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3.6.2.16 Electromagnetic compatibility. When specified (see 6.2), item 1, grinders shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.6.4).

3.6.3 Item 2, sizes 40, 60, and 100 mixer-grinder. The mixer-grinder shall consist of a mixing hopper, lid, and agitator assembly, grinder assembly, frame assembly, electric motor, electric controls and casters. All fitted surfaces shall be smooth and properly aligned. The mixer-grinder shall be assembled so the assembly of components is complete and of proper interrelationship. When tested as specified in 4.6.3.3 moving parts shall show no undue vibration, bearings or bushings as applicable shall not overheat, and controls shall function as intended.

3.6.3.1 Mixing hopper. The mixing hopper shall be fabricated of stainless steel specified in 3.4.1.1. The mixing hopper shall be supported to the main frame by mountings.

3.6.3.2 Lid. The lid for the mixing bowl shall be fabricated from stainless steel specified in 3.4.1.1. The lid shall be substantially constructed, reinforced or braced in such a manner that the lid fits the mixing bowl opening. Hinged lid shall be provided with means to hold the lid in the fully open position. The lid shall be provided with a unit that will secure the lid when closed.

3.6.3.3 Agitator assembly. The agitator assembly shall consist of an agitator shaft with mixing arms, shaft supports, seals, lubricated bushings or bearings and a sprocket unit or other means of driving the agitator shaft. The agitator shaft shall be fabricated of stainless steel specified in 3.4.1.2. The agitator shaft shall be supported at each end by shaft supports. The shaft supports shall be provided with either bearings, lubricated bushings, or of a self-lubricating type bushings. The shaft supports shall be secured to the mixing bowl ends. When other than sealed bearings are provided for the agitator shaft means shall be furnished for lubrication. The agitator shaft shall be removable for cleaning.

3.6.3.4 Grinder assembly. The grinder assembly shall consist of a feed worm housing, feed worm, knife, end-plate, clamping ring, thrust bearing and power transmission.

3.6.3.4.1 Feed worm housing. The feed worm housing shall conform to the requirements of 3.5.1 and shall be detachable from the mixing bowl.

3.6.3.4.2 Knife. The knife shall conform to the requirements of 3.5.3. Unless otherwise specified (see 6.2), three knives shall be provided for the mixer-grinder.

3.6.3.4.3 End-plates and clamping ring. End-plates and clamping ring shall conform to 3.5.4 and 3.5.5, respectively. Unless otherwise specified (see 6.2), end plates for the mixer-grinder shall conform to the size 30 grinder requirements specified in table III. Unless otherwise specified (see 6.2) height from floor to center to the end-plate shall be minimum 23 inches.

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3.6.3.4.4 Power transmission. The mixer-grinder shall be equipped with the necessary gears, gear shafts, chains, and sprockets for the transmission of power from the motor shaft to the grinder feed worm and to the mixer agitator shaft. Components shall be encased and securely fastened within the housing. Transmission bearings shall be located and arranged to prevent lubricating oil or grease from contacting the meat product. Gears, sprockets, and chains shall be of adequate load-bearing capacity to meet the tests specified in 4.6.3. Gears and sprockets shall be keyed or splined to the shafts.

3.6.3.5 Frame assembly. The frame assembly shall be constructed of formed or rolled steel angles specified in 3.4.5 or of pipe specified in 3.4.6. The frame assembly shall be constructed to support the mixer-grinder components. The frame assembly may be of the open frame type or of the inclosed frame type. Open type frame assembly shall be of a welded construction consisting of a welded bottom frame with four casters located at each of the four corners and support bracings welded at the bottom to the bottom frame and secured at the top to the mixing bowl by bolting or welding. Closed type frame assembly shall be of a welded construction with tubular uprights forming the legs and with horizontal cross bracing to support the frame bottom. Complete frame assembly to be galvanized dipped in accordance with manufacturer's standard practice, when applicable. Support bracing shall be provided for rigidity. The side panels may be secured to the frame by bolting or welding to the frame inclosing the frame and components therein. An access panel for component maintenance shall be provided.

3.6.3.5.1 Casters. Casters and caster mounting areas shall show no evidence of damage, distortion or failure when tested as specified in 4.6. Four casters shall be furnished, four swivel or two swivel and two rigid casters. The swiveling casters shall be located on the operator's side of the mixer-grinder machine. The swivel casters located on the operators side shall be provided with foot-operated brakes.

3.6.3.6 Motor. The mixer-grinder machines shall be provided with an electric motor with a minimum horsepower (h.p.) rating of 7-1/2 h.p. and shall conform to the requirements of 3.6.2.11.

3.6.3.7 Motor controls. The motor controls for the mixer-grinder machine shall conform to the requirements of 3.6.2.13.1 and to size 50 grinder requirements specified in table V and requirements of 3.6.2.13.2 through 3.6.2.16. The mixer-grinder machine shall be provided with means for mix-grind operation, mix only operation, and reversing operation. A safety interlock switch shall be provided for the mixing bowl lid and shall break the mixer-grinder electrical circuit when the mixing bowl lid is raised at least 1/8 inch from the closed position. When specified (see 6.2) a foot operated switch shall be provided.

3.6.3.8 Capacity. Mixer-grinder machines shall be capable of mixing and grinding a minimum of 40 pounds of beef on second grind through a 1/8 inch hole and enc plate for 60 hertz (Hz) a.c. system. Mixer-grinder machines with 25-50 Hz a.c. system shall have a rate not less than 32 pounds of beef on second grind. The ground beef fat content shall not exceed the per cent specified, after second grind, when tested (see 4.6.3.3).

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3.6.3.9 Lubrication. Components requiring lubrication shall conform to the requirements of 3.6.2.9.

3.6.4 Item 3 mixer-grinders. Item 3 mixer-grinders shall consist of two meat grinders connected together to form an integrated system. The item 3 mixer-grinders shall comply with the applicable requirements of 3.6.3 through 3.6.3.9. Item 3 mixer-grinders shall be motor operated and have a capacity of 100 pounds.

3.6.5 Item 4 chopping head assembly. Item 4 chopping head sizes shall be as specified in 1.2. Each size (see 6.2), chopping head assembly shall consist of a feed worm housing, feed worm, knife, end plate, and end plate clamping ring and shall comply with the applicable requirements of 3.6.2.3 through 3.6.2.5, respectively.

3.7 Identification marking. The item 1, meat grinders and item 2 mixer-grinder shall be permanently marked in accordance with MIL-STD-130 with the manufacturers name, tradename, or trademark, model and serial number of such known character as to be readily identifiable by the manufacturer. The nameplate for the grinders for shipboard use shall comply with type A, B, or C, of MIL-P-15024.

3.8 Repair and maintenance. Provisions shall be made to allow for adjustment, service and replacement of the knives, end plates, motors, electrical components, bearings, gears, sprockets, feed worm, and meat pan. In addition, there shall be free acess to inspect, service, and adjust any component or equipment without requiring disassembly of any other major component or equipment on the machine.

3.9 Finish. All exposed surfaces of the grinder shall be smooth on all surfaces and edges and shall be free from discoloration or stains.

3.9.1 Bowl of feed chute and worm housing. The bowl of feed chute and worm housing shall be tin-plated in accordance with the manufacturer's standard practice. The coating shall be adherent and free from blister, pits, or nodules. When an aluminum feed chute is provided, it shall be anodized in accordance with the manufacturer's standard practice. The mixing bowl shall be finished in accordance with the manufacturer's standard practice.

3.9.2 Other exposed surfaces. The motor and transmission housing shall be completely and uniformly coated with durable vitreous enamel or enamel paint. Vitreous enamel shall be applied over a ground coat and enamel paint shall be applied over a priming coat. Vitreous enamel shall be uniform and free from fish scaling, crazing, tearing, and copperheads. Enamel paint shall be free from drips, sags, runs, or chips. Unless otherwise specified (see 6.2), the color shall be in accordance with the manufacturer's standard practice. Stainless steel shall have a No. 4 or better standard polish.

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3.10 Workmanship. All components and assemblies of the meat grinder or mixer-grinder shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts and stampings, if used, shall be free of sand, fins, pits, blow holes and spruces. External surfaces shall be free from sharp edges and corners, except when corners are required.

3.10.1 Metal fabrication. Metal used in the fabrication of the meat grinder or grinder-mixer shall provide for original quality surface finish and shall be free from kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be made with dies or fixtures to insure uniformity of size and shape.

3.10.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts or incompleted fusion. All scale and flux shall be removed from the finish weld area.

3.10.3 Fastening devices. Threaded fastener and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked or stripped and shall be securely torqued. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required, item 1 meat grinder, item 2 or item 3 mixer-grinder or item 4 end plates, as applicable shall be examined for defects listed in table VI and dimensions specified, and tested as specified in 4.6.1, 4.6.2, and 4.6.3.1. In addition for item 1, type I grinders shall be tested in accordance with 4.6.3.1 and 4.6.5, as applicable. Item 1, type II grinders and item 2 or item 3 mixer-grinders shall be tested in accordance with 4.6.3, 4.6.3.2 thru 4.6.6, as applicable. Failure to pass any tests

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shall be cause for rejection of the first article. The presence of any defects listed in table VI or any dimension not within specified requirements shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined, and tested in accordance with referenced specification and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 Intermediate testing of knife, blades and end-plates. The knife blades and end-plates shall be tested for hardness in accordance with 4.6.2, using inspection level S-3 with an acceptable quality level (AQL) of 4.0 defects expressed in terms of defects per hundred units. The lot shall consist of all knives offered and all end-plates offered for inspection at one time. The sample unit shall be one knife and end-plate, of each size specified.

4.4.3 End item inspection. The inspection lot shall be all meat grinders of one type, size and style with accessories or all mixer-grinders offered for inspection at one time. The sample unit shall be one completely fabricated meat grinder with accessories or mixer-grinder.

4.4.3.1 Visual examination. The completely assembled grinder with specified accessories or mixer-grinder shall be examined for defects classified in table VI. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE VI. Classification of defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturers standard product except for changes necessary to comply with specification requirements	X	
Finish	Not as specified		X
	Tincoating contains blisters, pits, or nodules	X	
	Fish-scaling, crazing, tearing, or copper-heads (vitreous enamel)	X	
	Painted finish contains runs, drips or sags		X
	Evidence of discoloration, or stains	X	

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TABLE VI. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Construction and workmanship general	Component missing	X	
	Any wearing part not accessible for adjustment or replacement	X	
	Any component misplaced or not in alignment	X	
	Any burr, sliver, sharp edge, rough die, tool, gouge, trim, or grind marks	X	
	Any component fractured, buckled, bent, dented, punctured, or malformed	X	
	Functioning component that is inoperative or will not operate as intended	X	
	Functioning component that required abnormal force to operate		X
	Any adjustable assembly not properly adjusted to perform the intended function		X
	Not constructed as specified	X	
	Not fabricated of material specified	X	
Welding	Missing	X	
	Not free from rust scale, paint, grease and other foreign matter	X	
	Scale and flux deposits not removed		X
	Weld not smooth and free of cracks, burnholes, undercuts or incomplete fusion		X
Fastening devices	Missing, cracked, or stripped		X
	Fastener, loose		X
	Fastener improperly torqued	X	
	Rivet holes not accurately punched	X	
	Rivet not peened properly		X
Castings	Miscast, blowholes, or incomplete, not free of sand, fins, pits, and sprues	X	
Detailed construction feed worm	Feed worm housing not cast of one piece	X	
	Feed worm bearing surfaces not machined smoothly	X	
	Not provided with flutes as specified	X	

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TABLE VI. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Knives	Not specified number of blades	X	
	Surfaces designed for contact with end plates not flat and smooth	X	
End plates	Not notched or provided with flats or recesses for interlocking	X	
	Clamping ring excessively loose or binds on threads	X	
Thrust bearing	Not type specified (when bearing is provided)-	X	
	Not removable	X	
Item 1, type I grinders	Not type base specified i.e. (screw-clamp or slide-clamp)	X	
	Crank not detachable	X	
	Handcrank not provided with wooden, plastic, or metal handle	X	
	Not provided with correct amount of knives and end-plates		X
Item 1, type II grinders:			
Base plate	Supports not provided	X	
	Legs not provided (style 2)	X	
	Leg adjustment as specified (style 2)		X
Meat pan	Pan does not fit feed chute	X	
	Not specified material	X	
	Not firmly supported at housing		X
	Safety guard missing or not as specified	X	
Feed housing	Not detachable from meat pan	X	
Stomper	Not sized or shaped as specified		X
Item 2			
Mixer-grinder	Not as specified	X	
	Component missing	X	

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TABLE VI. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Mixing hopper	Not supported as specified	X	
Lid	Not as specified	X	
	Does not fit mixing bowl opening	X	
	Does not remain fully open	X	
Agitator	Not as specified	X	
	Not removable for cleaning	X	
Power transmission	Lubricating oil or grease contacting meat product	X	
	Components not encased	X	
	Components not fastened	X	
Frame assembly	Not as specified	X	
	Support bracing not secure	X	
	Maintenance access panel not provided	X	
Casters	Not as specified	X	
	Not secured	X	
	Swivel casters not provided with brakes	X	
Electrical assembly	Electric cord missing	X	
	Switches not recess type	X	
	Wiring cut, abraded, not properly joined, loose at terminals, or not inclosed in conduit where required	X	
	Adequate slack not provided for wiring to relieve strain or excessive insulation stripped from wiring	X	
Lubrication	Missing or not as specified	X	
Switches, starter, and other controls	Missing, not type specified, or do not function as intended	X	
Marking identification	Missing, incomplete, not legible, not corrosion-resistant metal	X	
Instructions	Missing, incomplete, not legible	X	

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alarm condition during the test period. Select alarm points so that

4.4.3.2 Dimensional examination. Item 1, meat grinder and specified accessories and item 2 mixer-grinder shall be examined for compliance with dimensional requirements specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.4.3.3 End product testing. When no first article is required, the initial production unit shall be tested as specified in 4.6.1, 4.6.2, and 4.6.3 and 4.6.3.1 for item 1, type I grinders and 4.6.2, 4.6.3, 4.6.4, and 4.6.6 as applicable, and 4.6.5 for item 1, type II grinders and item 2, and item 3 mixer-grinders.

4.4.4 Inspection of preparation for delivery. An examination shall be made to determine that preservation, packaging, packing, and marking as required by section 5 are complied with. Defects shall be scored as specified in table VII. The inspection level shall be S-4 with an AQL of 6.5 defects, expressed in terms of defects per hundred units. The sample unit shall be one shipping container fully prepared for delivery. The lot shall consist of the number of containers offered for delivery at one time.

TABLE VII. Examination of preparation for delivery

Examine	Defect
Markings, exterior and interior	Omitted; incorrect; illegible, improper size, location, sequence, or method of application.
Preservation	Preservative improperly applied or missing.
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as blocking and cushioning inadequate, improper, or missing.
Contents	Contents per container not as specified.

4.4.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-601, PPP-B-621, PPP-B-585, or MIL-C-104, examination for defects in closure and strapping shall be in accordance with applicable containers specification.

4.5 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

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4.6 Tests.

4.6.1 Strength test (Item 1, type 1). The item 1, type I grinder shall be clamped to a table. The clamp shall be tightened in accordance with the torque shown in table VIII and both the clamp and the body observed to determine conformance with the nondeforming requirement of 3.6.1.3. The moment for testing the body shall be applied through the feedworm hole provided for attachment of the handcrank. The moment shall be in the plane of the clamp and such a direction as to tend to open the clamp. The specified moment is the value about the centerline of the screw of the clamp.

TABLE VIII. Strength test (item 1, type I)

Grinder size	Clamping torque in foot-pounds	Moment in foot-pounds when the clamping torque is 10 foot-pounds
3	15	70
5	20	80
8	25	90
10	30	100

4.6.2 Rockwell hardness. The knives and end-plates shall be tested for hardness in accordance with ASTM E 18 to determine conformance with hardness requirements of 3.5.3 and 3.5.4. Hardness readings shall be taken as near as practicable to the cutting edge on the flat surface of each blade and on any area between perforations of the end plates. Inability to meet the hardness requirements specified shall constitute failure of this test.

4.6.3 Performance tests. The grinders shall be tested in accordance with 4.6.3.1, 4.6.3.2, or 4.6.3.3 as applicable Meat used for this test shall conform to the boned beef requirements of MIL-B-3854. For hand-operated machines, the pieces of meat shall be approximately 2-inch cubes or strips 2 by 2 inches in cross section, except that the meat for the size 3 machine shall be approximately 1-1/2-inch cubes or strips 1-1/2 by 1-1/2 inches in cross section. For electric-operated machines, the meat shall be cut in strips, approximately 2 by 2 inches in cross section. The electric grinder shall be connected to an electrical supply having the same characteristics specified. Any nonconformance with the requirements specified shall constitute failure of this test.

4.6.3.1 Item 1, type I. The test shall be conducted using a 3/16-inch-hole end-plate. The meat shall be fed continuously into the bowl hopper and the crank rotated for one hundred revolutions. The ground meat shall be weighed to determine conformance with the grinding rates specified in 3.6.1.6. Inability to meet the grinding rate specified shall constitute failure of this test.

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4.6.3.2 Item 1, type II. The test shall be conducted using a 1/8-inch-hole end-plate. The meat shall be fed continuously into the bowl or feed chute, using the stomper if necessary for a minimum period of 1 minute to determine conformance with the grinding rate specified in 3.6.2.10 for class and size of grinder specified. During the test the grinder shall be tested to determine conformance with 3.6.2 and 3.6.3. Any nonconformance with the requirements specified shall constitute failure of this test.

4.6.3.3 Item 2. The test shall be conducted using a 1/8-inch hole end-plate. The boned beef shall be fed continuously into the feed chute by the mixer bowl agitator for a period of 1 minute to determine conformance with the grinding rate specified in 3.6.3.8. The fat content of the ground beef shall not exceed the 22% when tested in accordance with the thermal extraction method of MIL-B-3854. During the test the mixer-grinder shall be tested to determine conformance with 3.6.3. Any nonconformance with the requirements specified shall constitute failure of this test.

4.6.4 Electromagnetic compatibility tests. When electromagnetic compatibility is required the first article or initial production unit, as applicable, shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461. Any noncompliance with the requirements specified shall constitute failure of this test.

4.6.5 Dielectric test. Apply a 60-cycle alternating potential between current carrying parts and any noncurrent-carrying metal parts for a period of one minute with the device at the operating temperature reached in normal use to determine compliance with 3.6.2.14. The test potential shall be 900 volts for a machine employing a motor rated at 1/2 horsepower or less, or rated for operation on 250 volts or less; or twice rated voltage plus 1,000 volts for a machine employing a motor rated at more than 1/2 horsepower or rated for operation on more than 250 volts. Capacitors, if used, shall be tested at the same potential as the machine with which they are employed. Any noncompliance with the requirements specified shall constitute failure of this test.

4.6.6 Obstacle. Item 2 mixer-grinder and item 3 mixer-grinder, as applicable, shall be loaded with an evenly distributed load of 500 pounds. The loaded mixer-grinder shall pass over a 1/4 inch obstacle ten times at a minimum speed of 2 miles per hour to determine compliance with 3.6.3.5.1. Any noncompliance with the requirements specified shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

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5.1.1 Level A. Preservation and packaging shall be in accordance with the method and material requirements of MIL-P-116 as hereinafter specified.

5.1.1.1 Preservation.

5.1.1.1.1 Cleaning and drying. All exposed, uncoated ferrous metal surfaces including corrosion-resisting steel surfaces of each grinder and mechanical accessory and repair parts, shall be cleaned process C-1 and thoroughly dried.

5.1.1.1.2 Application of preservative. All cleaned and dried surfaces shall be coated with type P-14 preservative. Mechanical spare parts and preserved surfaces of each grinder, which might contact packaging and packing materials, shall be wrapped or covered with barrier-material conforming to type II, grade A of MIL-B-121. Barrier material shall be secured with tape conforming to type III, grade A of PPP-T-45 or PPP-T-60, tape or twine, or heat sealed when heat sealable barrier material is used.

5.1.1.1.3 Lubrication. All bearings shall be lubricated to capacity with the manufacturer's recommended lubricant containing a rust inhibitor.

5.1.1.2 Packaging.

5.1.1.2.1 Item 1, type I grinders Prior to packaging, the handcrank and handle, or handle only, shall be removed, or removed and reassembled to the grinder so as to reduce the cube of the package. Knives and end-plates not installed in the grinder, preserved as specified in 5.1.1.1.2, shall be packaged method III. Each type I grinder, together with extra knives and end-plates, and disassembled components, preserved and packaged as specified shall then be packaged in a fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. The contents shall be cushioned and immobilized within the box to preclude damage while in transit. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.1.1.2.2 Item 1, type II grinders. The four rubber supports or legs, when removable, shall be detached from the base plate of each type II grinder. Knives and end-plates not installed in the grinder, preserved as specified in 5.1.1.1.2, shall be packaged method III. The meat stomper when not attached to the machine, rubber supports or legs, and knives and end-plates shall then be packaged in a fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. The contents shall be cushioned and immobilized within the box to preclude damage while in transit. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.1.1.2.3 Item 2, and 3 mixer-grinder. All detachable components which will reduce cube or might become damaged while in transit, shall be disassembled from the mixer-grinder and packaged method III.

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5.1.2 Level C. Each grinder or mixer-grinder shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Item 1, type I grinder and item 4. Item 1, type I grinders or item 4 of one size only, preserved and packaged as specified in 5.1, shall be packed in the quantity specified in table IX in a snug-fitting shipping container conforming to overseas type of PPP-B-601; class II of PPP-B-591; class 2 of PPP-B-576; class 2, style 2 or 4 of PPP-B-621; style 1 or 2, class 3 of PPP-B-585; or class 2, grade A, style E of PPP-B-640. Each shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

TABLE IX. Packing of item 1, type I and item 4 machine

Size	Quantity per shipping container
3	12
5	6
8	4
10	3
42	2
46	2
52	2

5.2.1.2 Item 1, type II grinder and item 2, and 3 mixer-grinder. Each type II grinder or mixer-grinder, preserved and packaged as specified in 5.1, shall be packed in a shipping container conforming to overseas type, type 3 load of PPP-B-601; class 2, style 2 or 4, type 3 load of PPP-B-621; or type I, class 1 or 2, style a of MIL-C-104. Each shipping container conforming to PPP-B-601 or PPP-B-621, regardless of gross weight, shall be provided with skids in accordance with the applicable container specification, and the contents shall be covered with a full-height shroud made of waterproof barrier material conforming to class E-2, H-1, H-5, or M-1 of PPP-B-1055. All seams of the shroud shall be sealed with a 3/4-inch minimum width continuous seam of water-resistant adhesive conforming to MIL-A-260. Shipping containers conforming to MIL-C-104 shall be used when weight of contents exceeds 1000 pounds. The fiberboard box containing accessories, shall be secured in an unused portion of the shipping container. Contents shall be anchored in accordance with the appendix of MIL-C-104. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

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5.2.2 Level B.

5.2.2.1 Item 1, type I grinder. Type I grinders of one size only, preserved and packaged as specified in 5.1, shall be packed in the quantity specified in table IX, in a snug-fitting container conforming to domestic type, style A or B or PPP-B-601; class 1, style 2 or 4 of PPP-B-621; style 1 or 2, class 1 of PPP-B-585; or class 2, grade A, style E of PPP-B-640. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.2.2 Item 1, type II grinder and item 2, and 3 mixer-grinder. Each item 1, type II grinder or each item 2, mixer-grinder preserved and packaged as specified in 5.1, shall be packed as specified in 5.2.1.2, except that shipping containers shall conform to domestic type, style A or B or PPP-B-601; class 1, style 2 or 4, type 3 load of PPP-B-621; or type I, class 1 or 2, style a of MIL-C-104; waterproof shroud and waterproof liner for container conforming to MIL-C-104, shall not be required.

5.2.3 Level C. Grinders, and mixer-grinders preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The item 1, grinders covered herein are intended for use in kitchens, galleys, central meat cutting plants, and commissary store meat departments. Items 2, and 3, mixer-grinders are intended for use in central meat cutting plants. Item 4 is used to increase the grinding load capacity.

6.1.1 Item 1, type I. The size 3 grinder is intended for use where the personnel to be served does not exceed ten individuals. Sizes 5 and 8 grinders are for general services. Size 10 grinders may require the services of two men to maintain continual operation.

6.1.2 Item 1, type II. The following guide based on a meat allowance of 40 pounds per one hundred individuals for a single meal may be used in determining size of grinder required;

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documented ex-

Data is included in the chain. If the commander ... els described by standards and

Size of grinder	Number of individuals
4	Up to 600
8	601 to 1200
16	1201 to 2400

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Item, type, size, and style of grinder and size of mixer-grinder required (see 1.2).
(c) When a first article is required (see 3.1).
(d) Quantity of knives, if other than specified (see 3.6.1.1, 3.6.2.4 and 3.6.3.4.2).
(e) Quantity and size of end-plates, if other than specified (see 3.6.1.2 and 3.6.2.5 and 3.6.3.4.3).
(f) When end-plates having elliptical slots are required for size 3 grinder and the quantity (see 3.6.1.2).
(g) Whether a slide-clamp base or a screwdown base is required in lieu of a screw-clamp base (see 3.6.1.3).
(h) When floor bolting is required (see 3.6.2.2).
(i) Height to end plate center if other than specified (see 3.6.2.2, and 3.6.3.4.3).
(j) Electrical characteristics (see 3.6.2.11).
(k) Wiring, if other than specified (see 3.6.2.14).
(l) Whether fungus proofing is required (see 3.6.2.15).
(m) When electromagnetic compatibility is required (see 3.6.2.16).
(n) When a foot switch is required (see 3.6.3.7).
(o) Size of chopping head required (see 3.6.5).
(p) Color, if other than specified (see 3.9.2).
(q) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
(r) Marking required (see 5.3).

6.3 Level B packaging. When level B preservation and packaging is specified for civil agency procurement, the requirements of 5.1.1, level A preservation and packaging shall apply.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item, or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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6.5 Contract data requirement. Any requirements for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - MD
Navy - SH, MC, SA

User activities:

Army - CE

Preparing activity:

Army - GL

Project No. 7320-0400

Civil Agency Coordinating Activities:

GSA-FSS
VA-DMS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION 00-G-667F
Grinders, Meat, Electric; Grinders, Meat, Hand; And Mixer-Grinder, Meat, Electric

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

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